

Improving Transient Stability of Photovoltaic-Hydro Microgrids Using Virtual Synchronous Machines

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Abstract— Transient stability of photovoltaic-hydro microgrid systems is poor due to lack of inertia and the intermittent nature of photovoltaic systems. The stability of such systems can be improved by using virtual synchronous machines which add inertia into the system to allow for higher PV penetration without losing stability. A PV-hydro system was simulated in MATLAB/Simulink to analyze the transient stability problems due to PV fluctuations and/or load or generation changes. Virtual synchronous machine was added which reduced the frequency deviations and the high rate of change of frequency. The results indicate that the transient stability of photovoltaic-hydro microgrid systems can be improved by using virtual synchronous machines using a minimum amount of energy storage. However, the power requirements of the virtual synchronous machine's converter was found to be high compared to the overall system size. Furthermore, high PV penetration levels were achieved by adding virtual synchronous machine to the system.

Index Terms—Energy storage, Hydroelectric power generation, Photovoltaic systems, Power system stability

I. INTRODUCTION

Photovoltaic (PV)-hydro hybrid microgrid systems are attractive in remote regions rich in hydro resources and where the electric grid has not expanded. PV-hydro integrated systems can improve the reliability of PV systems without the need for expensive and hazardous storage devices. PV plants are also a perfect complement to small hydro systems during dry seasons when power production from the hydro plant is low [1]. These systems can work together for the socio-economic development of rural communities in developing countries like Nepal, India, Bhutan etc. rich in hydro resources [2]. At the same time, PV-hydro systems provide accessibility to sustainable and reliable energy at reduced costs. However, hydro plants have limited inertia and damping properties while PV systems are void of any inertia and damping properties [1]. Increased PV penetration in these PV-hydro integrated systems can have a significant impact on its dynamic performance and transient stability, especially in the isolated mode of operation due to the lack of inertia and damping from conventional synchronous generators on the grid. Load/generation changes and PV

fluctuations in such systems can cause large frequency deviations at a high rate of change of frequency (ROCOF) [3]. A solution to improve the transient stability of such integrated PV-hydro microgrids is to add inertia and damping into the system using virtual synchronous machines (VSM) [4].

The integration of PV systems with hydro system improves the characteristics of both sources. Moreover the cost of such systems is comparable to other renewable energy systems [5], but only 45% penetration level can be achieved if the frequency of system is to stay within the normal operating range [6]. The ISO standard 8528-5 for generating sets defines a normal operating range of $\pm 2.5\%$ for the frequency and recommends that the frequency should not exceed the $\pm 15\%$ critical operating range. Furthermore, the recommended recovery time if the frequency goes of the normal operating range is 10 seconds. The maximum recommended ROCOF value is $\pm 1\%$ of the rated system frequency [7]. The operation of generators outside the recommended levels can lead to degraded generator performance. Moreover, sustained periods of frequency variations affects proper operation of frequency dependent loads, especially electric motors [8].

Whenever there is a frequency variation in the system, the inertial reserve of the power system acts to counter act the initial frequency deviation before the primary reserve brings the frequency back to a steady state value [9]. Conventionally, the large rotational masses of synchronous generators are responsible for providing a significant amount of inertia to large power systems. Synchronous generators release or absorb kinetic energy whenever there is a change in grid frequency [10]. But with a high penetration of PV, the overall inertia of the system decreases in proportion to the increase in system size. A PV- hydro system will exhibit fast frequency dynamics which cannot be handled by the conventional governors of hydro systems, responsible for the primary frequency control of the system. Large frequency deviations at high ROCOF thus may occur in such systems tripping frequency relays and causing unnecessary load shedding [6]. Hence, additional inertial response is needed to provide better dynamic frequency support during the transient periods of frequency deviations.

The case study was performed to develop PV–hydro microgrid systems in a country like Nepal where numerous hydro systems have been proposed but have not been realized due to the absence of transmission lines and/or unsustainability of these systems operating autonomously due to long dry

periods. The benchmark of the PV-hydro microgrid system developed consists of a 39 kW hydro and 25 kWp PV as shown in Fig. 2. The microgrid is a three phase system operating at 208 V (line to line) with a rated frequency of 60 Hz. The size of PV model was chosen to match the PV system size of the microgrid lab in South Dakota State University (SDSU). Similarly, the hydro model was sized to match generator capacity. These values were used for future experimental analysis of the simulations in this paper. The hydro system was modeled in MATLAB/Simulink using a three phase synchronous generator along with its appropriate excitation system and an isochronous governor. The inertia constant, H , of the hydro system was assumed to be 2 s which is a typical value for hydro systems [17] and the PID settings of the isochronous governor used are 3.50, 0.54 and 1.06 respectively. The PV system was modeled using three current sources, with no inertial response. An inverter was not used as a model for the PV system to reduce the simulation overhead. The output power from the PV system thus depended on the solar irradiance data fed into the model obtained from the data acquisition system in the microgrid lab at SDSU as discussed later. The PV model based on solar irradiance determined the current that was injected into the microgrid. The electrical load used in the case study was purely resistive.

The frequency variations in the benchmark PV-hydro microgrid system was tested for two cases, first a step irradiance change from 750 W/m² to 250 W/m² and second for a 250 second snapshot of real irradiance data obtained from the data acquisition system in the microgrid lab as shown in Fig. 3. The irradiance data was sampled once per second. This particular snapshot of the irradiance data was used for this study because there are large variations in the irradiance data as well as instances of constant irradiance which allowed for transient analysis of the system under different conditions.

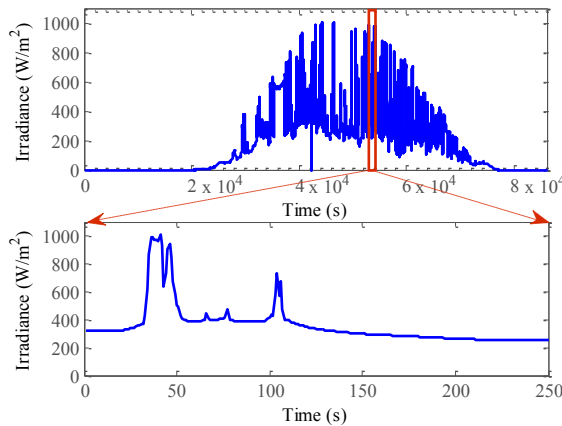


Figure 3. 250 second snapshot of real irradiance data obtained from the microgrid lab at SDSU from July 19, 2012. (Sampling period = 1s)

B. Virtual Synchronous Machines Design and Analysis

The details of the virtual synchronous machine implemented in the benchmark PV-hydro microgrid system is shown in Fig. 4. A phase locked loop (PLL) was used to measure changes in system frequency and the ROCOF. Based on these measurements the power ' P_{VSM} ' that the VSM needed to exchange with the microgrid system was calculated using (1). The inertial constant ' K_I ' and the damping constant ' K_P ' were

calculated using (2) and (3) respectively and are shown in (5) and (6). The nominal power rating of the VSM, ' P_{VSM_NOM} ' was taken to be 6400 W which was 10% of the total size of the benchmark PV – hydro system. Using ISO standards, the maximum allowable frequency deviation was 3 Hz in either direction ($\pm 2.5\%$) and maximum allowable ROCOF was 1.2 Hz/s in either direction ($\pm 1\%$) [7]. The values of ' $ROCOF_{max}$ ' and ' Δf_{max} ' were selected to be slightly lower at 0.5 Hz/s and 2 Hz respectively. An additional feedback signal of the change in state of charge of the energy storage device ' ΔSOC ' was also added to maintain the battery state of charge at 50%. This ensures that the VSM is always ready to either inject or absorb power. The energy storage device either absorbed or injected power depending upon whether the state of charge was below or above 50%. Hence the modified equation to calculate the VSM power consisted of three components, the inertial component, the damping component and the state of charge component respectively as in (4). The value of ' K_{SOC} ' was selected such that VSM exchanged a maximum power of 2500 W when the SOC deviation was 30% in either direction as shown in (7). A higher or a lower value of ' K_{SOC} ' affects the rate at which the energy storage device was charged or discharged when the SOC deviated from desired value of 50%.

$$P_{VSM} = K_I \frac{d(\Delta f)}{dt} + K_P (\Delta f) + K_{SOC} (\Delta SOC) \quad (4)$$

$$K_I = \frac{6400 \text{ W}}{0.5 \text{ Hz/s}} = 12800 \quad (5)$$

$$K_P = \frac{6400 \text{ W}}{2 \text{ Hz}} = 3200 \quad (6)$$

$$K_{SOC} = \frac{2500 \text{ W}}{0.3} = 8333.33 \quad (7)$$

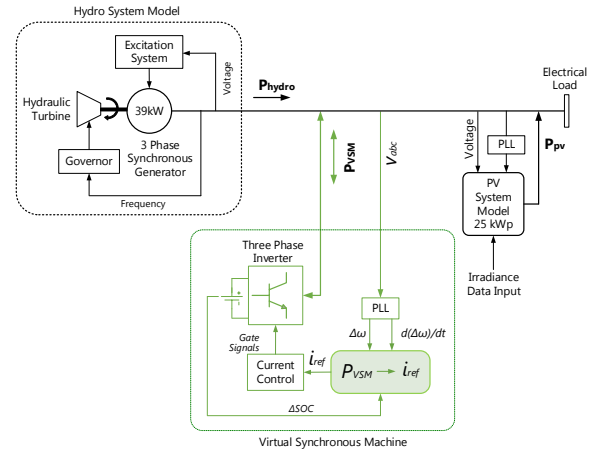


Figure 4. Details of the virtual synchronous machine implemented in the benchmark PV-hydro microgrid system.

The reference current ' I_{ref} ' for the current controller was set using the calculated power ' P_{VSM} '. Based on the reference current the current controller implemented a PI based control to generate the necessary gate signals for the three phase inverter. PSIM simulation software was used for the design and verification of the PI current controller. Based on the change in frequency and ROCOF, power was injected or/ absorbed by the VSM through the three phase inverter to emulate the inertial and damping properties of a real machine. The reduction in

frequency variations using the VSM in the PV-hydro microgrid was then studied using OPAL-RT real time digital simulation system which performs software in the loop simulations of the designed system. With the addition of the VSM the analysis was again performed for two cases: first, a step change in irradiance from 750 W/m^2 to 250 W/m^2 and second, for a 250 second of real irradiance data as defined before. The frequency variations in the PV-hydro system with and without the VSM were then compared. The requirements of the power converter for the VSM and its energy storage requirements were also studied.

IV. RESULTS AND ANALYSIS

A. Case A : Transient Analysis of PV-Hydro System with Step Change in Irradiance

To study transient behavior of the PV-hydro system with irradiance changes, the irradiance of PV model was changed from a constant value of 750 W/m^2 to 250 W/m^2 at a simulation time of 100 seconds as shown in Fig. 5(a). This was equivalent to an additional load of 19.5 kW for the hydro system. The load on the system was kept constant at 25 kW to ensure maximum penetration of 25 kWp PV system. The frequency variations of the system are shown in Fig. 5(b) by dotted lines. The frequency was well outside the normal operating range of $\pm 2.5\%$ but stayed within the critical operating range of $\pm 15\%$. The period for which the frequency was outside the normal operating range was 14 seconds, which was over the recommended limit of 10 seconds. More importantly, the ROCOF was very high at 8.2 Hz/s which was well over the maximum limit of 0.6 Hz/s as shown in Fig. 5(c). This was due to the small size of the system, and could result in tripping of the frequency relays, causing power production to stop.

Next, the VSM was added to the system for the same step irradiance change from 750 W/m^2 to 250 W/m^2 . The inertial and damping constant for the VSM were set at 3200 and 12800 as calculated in section III. The reduction in the frequency variations and ROCOF with the VSM are shown by solid lines in Fig. 5(b) and 5(c) respectively. The minimum frequency deviation in the benchmark PV-hydro system was reduced from 51.69 Hz to 59.15 Hz which was within the normal operating range of $\pm 2.5\%$. There was no overshoot of the frequency over the rated frequency of 60 Hz. Similarly, the ROCOF was reduced from 8.2 Hz/s to 0.6 Hz/s. This reduction should prevent unnecessary tripping of frequency relays. The addition of VSM reduced the minimum frequency excursion by 12.6 % and the ROCOF by 92.6%.

B. Case B: Transient Analysis of PV-Hydro System with Real Irradiance Data

To study the transient behavior of the PV-hydro system with real irradiance changes, a snapshot of 250 seconds of real irradiance data as described in section III was fed into the PV model as shown in Fig. 6(a). Initially, the simulation was performed without the VSM. The load on the system was kept constant at 25 kW to ensure maximum penetration of the PV system. The frequency variations of the system are shown in Fig. 6(b) by dotted lines. The frequency was well outside the normal operating range of $\pm 2.5\%$ but stayed within the critical operating range of $\pm 15\%$. There were three instances where the frequency went out of the normal operating range. In the second

instance, the recovery time of the frequency was 10.37 seconds which was marginally over the 10 second permissible limit. But more critically it was the ROCOF in the system which compromised the stability of the system. The maximum ROCOF was 4.8 Hz/s, well outside the 0.6 Hz/s limit.

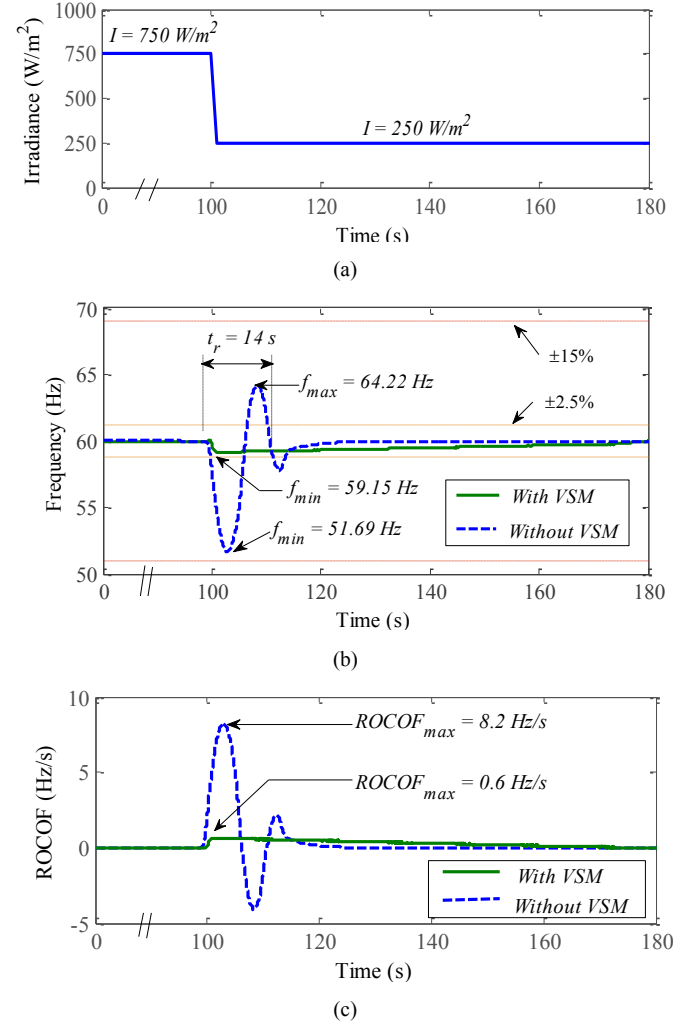


Figure 5. (a) Step change in input irradiance for the 25 kWp PV model from 750 W/m^2 to 250 W/m^2 . (b) Frequency variations in the benchmark PV-hydro system with and without VSM due to step change in irradiance (c) ROCOF in the benchmark PV-hydro system with and without VSM due to step change in irradiance

The VSM was then added to the system for the same irradiance pattern. The inertial and damping constant for the VSM were set at 3200 and 12800 as calculated in section III. The reduction in the frequency variations and ROCOF with the VSM are shown by solid lines in Fig. 6(b) and 6(c) respectively. The maximum frequency excursion was reduced by 6.3% and the minimum frequency excursion was reduced by 4.7%. Similarly, the maximum value of the ROCOF was reduced by 85.4%. The reduction in frequency variations and the ROCOF for different PV systems sizes of 10 kWp, 15 kWp and 25 kWp are summarized in Table I. In all cases, without the VSM the frequency and the ROCOF was outside the permissible limits. But with the addition of the VSM, the frequency and ROCOF are within the permissible limits. The addition of VSM makes integration of PV system possible while maintaining stability.

TABLE I. REDUCTION IN FREQUENCY VARIATIONS AND ROCOF OF THE PV – HYDRO SYSTEM WITH AND WITHOUT VSM FOR VARIOUS PV SYSTEM SIZES

PV System Size	Without VSM			With VSM		
	Maximum Frequency	Minimum Frequency	Maximum ROCOF	Maximum Frequency	Minimum Frequency	Maximum ROCOF
10 kWp	61.8 Hz	58.7 Hz	1.6 Hz/s	60.5 Hz	59.7 Hz	0.2 Hz/s
15 kWp	62.8 Hz	58.1 Hz	2.5 Hz/s	60.7 Hz	59.7 Hz	0.4 Hz/s
25 kWp	65.1 Hz	56.9 Hz	4.8 Hz/s	61.0 Hz	59.6 Hz	0.7 Hz/s

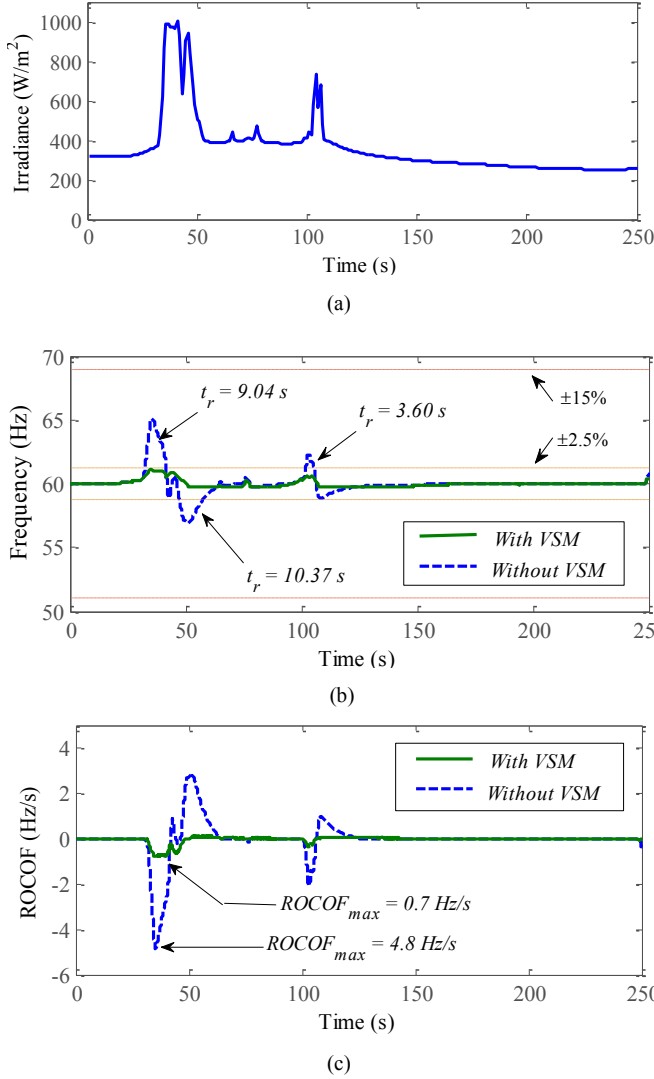


Figure 6. (a) 250 second snapshot of real PV irradiance data (b) Frequency variations in the benchmark PV – hydro system with and without VSM due to real irradiance changes (c) ROCOF in the benchmark PV-hydro system with and without VSM due to real irradiance changes

C. Case C : VSM Power and Energy Requirements

Fig. 7(a) shows the power that was exchanged by the VSM with benchmark PV-hydro system for a step change in irradiance from 750 W/m² to 250 W/m². The peak power that the VSM converter needed to handle was quite high at 10.1 kW. The contribution of the inertial term, the damping term and the state of charge terms to VSM power are shown in Fig. 7(b). When there was drop in irradiance the VSM provided dynamic

frequency support by injecting inertial peak power of 8.24 kW and damping peak power of 2.06 kW into the system. The power injection then gradually lowered as the frequency of the system came back to the rated frequency of 60 Hz. At the same time, since the energy storage device started discharging, the SOC component of VSM absorbed power from the system to charge the energy storage device. The peak power that SOC component absorbed from the system was 3.69 kW. As the SOC returned to 50% the power absorbed also decreased gradually. Similarly, Fig. 8 (a) shows the power that the VSM exchanges with the benchmark PV-hydro system when real irradiance data was used as described before. The peak power that the VSM handled in this case was 12.39 kW. This was higher than for the case of step load change in irradiance of 750 W/m² to 250 W/m² (change of 500 W/m²) because the real irradiance had irradiance changes higher than 500 W/m². In this case the inertial, damping and SOC component of VSM peak power were -10.21 kW, -2.56 kW and 1.94 kW respectively.

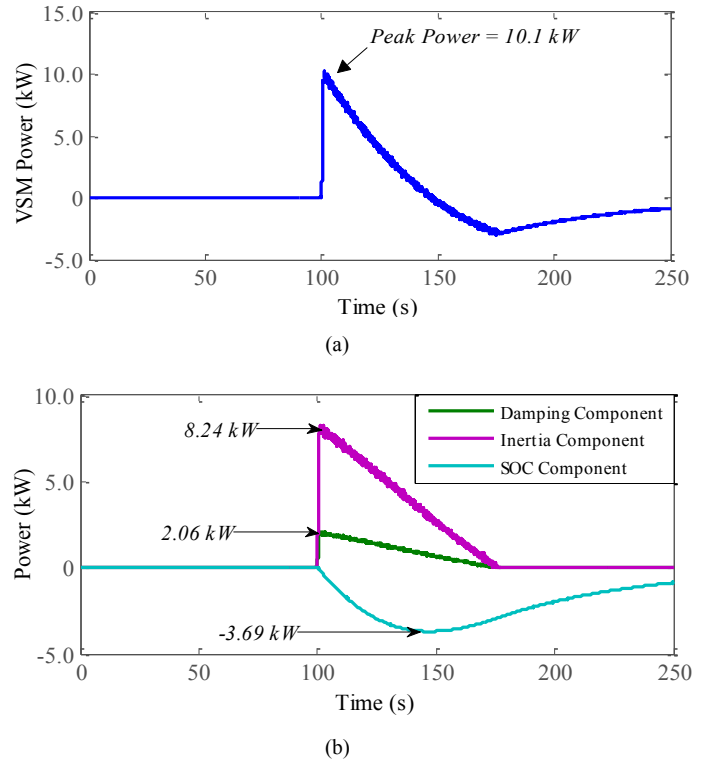


Figure 7. (a) Power flow through VSM for the test case of step change in irradiance from 750 W/m² to 250 W/m² (b) Proportional and derivate components of power flow through the VSM and battery charging/ discharging power for the test case of step change in irradiance from 750 W/m² to 250 W/m²

The net energy that the VSM exchanged with the microgrid was 11.7 Wh for the first case with the step change in irradiance from 750W/m² to 250 W/m². This was very low compared to the overall system size. Similarly, for the second case, with the real irradiance, the net energy exchange was negative at - 1.11 Wh. The energy storage requirements for different PV system sizes of 10 kWp, 15 kWp and 25 kWp along with the peak power requirements of the converter are summarized in Table II. It is evident that irrespective size of the PV system the energy storage requirement for the VSM is minimal. However, the power handling capacity requirement of the VSM increases significantly with the increase in PV system size.

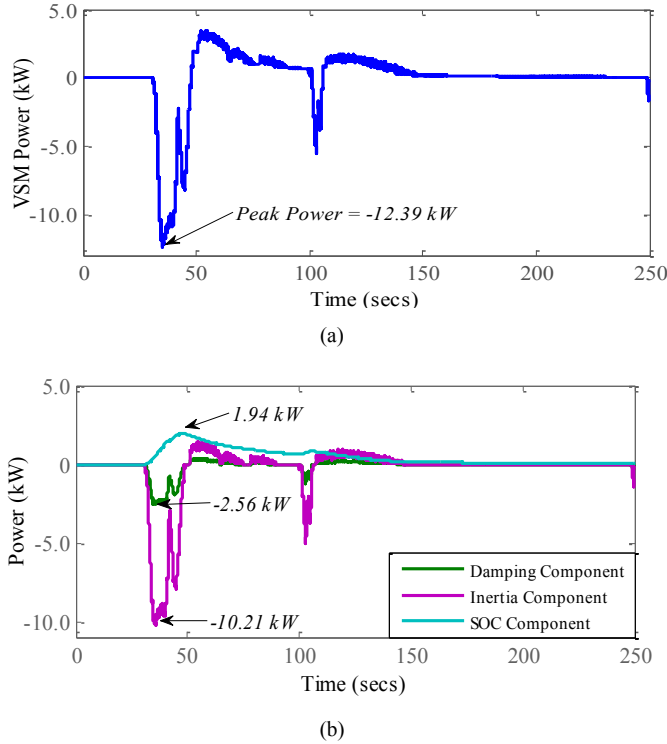


Figure 8. (a) Power flow through VSM for the test case with real irradiance data (b) Proportional and derivate components of power flow through the VSM and battery charging/ discharging power for the test case of real irradiance data

TABLE II. NET ENERGY EXCHANGE BETWEEN VSM AND BENCHMARK PV-HYDRO SYSTEM AND VSM PEAK POWER

PV System Size	Net Energy Exchange	VSM Peak Power
10 kWp	+ 0.04 Wh	4.51 kW
15 kWp	-0.34 Wh	7.19 kW
25 kWp	-1.11 Wh	10.21 kW

V. CONCLUSION AND FUTURE WORK

The feasibility of using virtual synchronous machines to improve the transient stability of PV-hydro microgrid systems was presented. High PV penetration can be achieved by emulating inertial requirements using VSMs. The frequency variations and the high rate of change of frequency due to either irradiance or load changes can be reduced to within desired limits using VSMs. The energy storage requirements of the

VSM are very low compared to system size. This indicates the possibility of using short term energy storage elements such as super-capacitors or the capacitance of the PV system instead of conventional storage devices. However, the power handling requirements of the converters used in the VSM are high which can add to the initial cost of the system. Future work should include study of implementing VSM using the capacitance of PV modules itself without using a dedicated energy storage element. Also, self-optimizing VSM algorithms can developed to reduce the power requirements of the VSM.

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